ZHIQI HUANG

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RESEARCH INTEREST

Information retrieval, natural language processing, data mining, and machine learning.

EDUCATION

University of Massachusetts, Amherst *June 2024*
Ph.D. in Computer Science
Advisor: Prof. James Allan

University of Maryland, College Park *May 2015*
M.A. in Statistics

Sun Yat-sen University *June 2013*
B.S. in Applied Mathematics

WORK EXPERIENCE

Capital One Jul. 2024 - Present
Applied Researcher *Boston, MA, USA.*

Work at the AI Foundations team, developing generative AI applications for financial use cases, with a focus on large language model (LLM) customization and retrieval-augmented generation (RAG).

Google Oct. 2023 - Jan. 2024
Research Intern *New York, NY, USA.*

Work on a project related to retrieval-augmented automatic speech recognition (ASR).

National Engineering Lab, Shenzhen University Jun. 2017 - Jul. 2018
Research Fellow *Shenzhen, China*

Work on uncertainty modeling and method for inductive learning.

Quad2Media Jul. 2015 - Mar. 2017
Data Scientist *Boston, MA, USA*

Design and manage a large-scale user profile system for the marketing campaign.

PUBLICATIONS

[1] **VerTox: Verifiable Reward-Guided Corpus Poisoning Against Neural Ranking Models**
Zhiqi Huang, Vivek Datla, Zhichao Xu, Puxuan Yu, Vivek Srikumar, Alfy Samuel. The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.

[2] **Learning to Route Queries to Heads for Attention-based Re-ranking with LLMs**
Yuxing Tian, Fengran Mo, Zhiqi Huang, Weixu Zhang, Jian-Yun Nie. The 49th ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2026.

[3] **Uncertainty as Feature Gaps: Epistemic Uncertainty Quantification of LLMs in Contextual Question-Answering**
Yavuz Bakman, Sungmin Kang, Zhiqi Huang, Duygu Nur Yaldiz, Catarina G. Belém, Chenyang Zhu, Anoop Kumar, Alfy Samuel, Salman Avestimehr, Daben Liu, Sai Praneeth Karimireddy. The 14th International Conference on Learning Representations (ICLR), 2026.

[4] **A Survey of Model Architectures in Information Retrieval**
Zhichao Xu, Fengran Mo, Zhiqi Huang, Crystina Zhang, Puxuan Yu, Bei Wang, Jimmy Lin, Vivek Srikumar. Transactions on Machine Learning Research (TMLR), 2026.

[5] **Distillation versus Contrastive Learning: How to Train Your Rerankers**
Zhichao Xu, Zhiqi Huang, Shengyao Zhuang, and Vivek Srikumar. The 4th Asia-Pacific Chapter of the Association for Computational Linguistics (AACL), 2025.

[6] **Confidence-Based Response Abstinence: Improving LLM Trustworthiness via Activation-Based Uncertainty Estimation**
Zhiqi Huang*, Vivek Datla*, Chenyang Zhu, Alfy Samuel, Daben Liu, Anoop Kumar, and Ritesh Soni. The 2nd Uncertainty-Aware NLP Workshop at EMNLP 2025.

[7] **An Automatic Method to Estimate Correctness of RAG**
Chi Zhang, Vivek V. Datla, Aditya Shrivastava, Alfy Samuel, Zhiqi Huang, Anoop Kumar, and Daben Liu. The 31st International Conference on Computational Linguistics (COLING), 2025.

[8] **Language Concept Erasure for Language-invariant Dense Retrieval**
Zhiqi Huang, Puxuan Yu, Shauli Ravfogel, James Allan. The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.

[9] **Optimizing Large-Scale Context Retrieval for End-to-End ASR**
Zhiqi Huang, Diamantino Caseiro, Kandarp Joshi, Christopher Li, Pat Rondon, Zelin Wu, Petr Zadrazil, Lillian Zhou. The 25th Interspeech Conference (INTERSPEECH), 2024.

[10] **UMass at TREC 2023 NeuCLIR Track**
Zhiqi Huang, Puxuan Yu, James Allan. The 32nd Text REtrieval Conference (TREC), 2023.

[11] **Search Result Diversification Using Query Aspects as Bottlenecks**
Puxuan Yu, Razieh Negin Rahimi, Zhiqi Huang, James Allan. The 32th ACM International Conference on Information and Knowledge Management (CIKM), 2023.

[12] **Soft Prompt Decoding for Multilingual Dense Retrieval**
Zhiqi Huang, Hansi Zeng, Hamed Zamani, James Allan. The 46th ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2023.

[13] **Hierarchical Transformer-based Query by Multiple Documents**
Zhiqi Huang*, Shahrzad Naseri*, Hamed Bonab, Sheikh Muhammad Sarwar, James Allan. The 13th International Conference on the Theory of Information Retrieval (ICTIR), 2023. *Best Paper Honorable Mention.*

[14] **Improving Cross-lingual Information Retrieval on Low-Resource Languages via Optimal Transport Distillation**
Zhiqi Huang, Puxuan Yu, James Allan. The 16th ACM International Conference on Web Search and Data Mining (WSDM), 2023.

[15] **Mixed Attention Transformer for Leveraging Word-Level Knowledge to Neural Cross-Lingual Information Retrieval**
Zhiqi Huang, Hamed Bonab, Sheikh Muhammad Sarwar, Razieh Negin Rahimi, James Allan. The 30th ACM International Conference on Information and Knowledge Management (CIKM), 2021.

[16] **AutoName: A Corpus-Based Set Naming Framework**
Zhiqi Huang, Razieh Negin Rahimi, Puxuan Yu, Jingbo Shang, James Allan. The 44th ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2021.

[17] **Learning to Rank Entities for Set Expansion from Unstructured Data**
Puxuan Yu, Razieh Negin Rahimi, Zhiqi Huang, James Allan. The 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2020.

[18] **Corpus-based Set Expansion with Lexico-syntactic Features and Distributed Representations**
Puxuan Yu, Zhiqi Huang, Razieh Negin Rahimi, James Allan. The 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2019.

[19] **PSynDB: Accurate and Accessible Private Data Generation**
Zhiqi Huang, Ryan McKenna, Gerome Miklau, Michael Hay, Ashwin Machanavajjhala. The 45th International Conference on Very Large Data Bases (VLDB), 2019.

(∗equal contribution)

RESEARCH PROJECTS

During my Ph.D., I participated in two projects funded by Intelligence Advanced Research Projects Activity (IARPA). My main contribution is designing models for specific retrieval tasks.

- Machine Translation for English Retrieval of Information in Any Language (MATERIAL).
  `https://www.iarpa.gov/research-programs/material`
- Better Extraction from Text Towards Enhanced Retrieval (BETTER).
  `https://www.iarpa.gov/research-programs/better`

AWARDS

Best Paper Honorable Mention 2023
ACM ICTIR

Dissertation Writing Fellowship 2023 and 2024
CICS, UMass Amherst

Winner of WSDM Cup 2023 2023
Second prize on multilingual information retrieval (MIRACL) challenge

PROFESSIONAL SERVICES

- Reviewer of ACM Special Interest Group on Information Retrieval (SIGIR).
- Reviewer of ACM International Conference on Information and Knowledge Management (CIKM).
- Reviewer of ACM International World Wide Web Conference (WWW).
- Reviewer of ACM International Conference on Web Search and Data Mining (WSDM).
- Reviewer of ACL Rolling Review System (ARR).
- Reviewer of IEEE Transactions on Knowledge and Data Engineering (TKDE).
- Reviewer of IEEE International Conference on Big Data (IEEE BigData).

MENTORSHIP

Mentor of Capital One Applied Research Internship Program (ARIP).

Teaching assistant for COMPSCI 240 Reasoning Under Uncertainty, Spring 2020; COMPSCI 689 Machine Learning, Fall 2019; COMPSCI 356 Digital Forensics, Spring 2019, UMass Amherst.